Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad,Natore

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Sessional Plan

Program:	Computer Science and Engineering (CSE)
Course Code & Title:	**CSE-2108 (Digital Logic Design Laboratory)**
Semester:	Summer-2025
Level:	2nd Year 1st Semester
Credit Hour:	1.50 Hr.
Name & Designation of Teacher(s):	**Must. Asma Yasmin** Associate Professor, Dept. of CSE
	Md. Saifuddin Lecturer, Dept. of CSE
Class Hours:	Wednesday: 11:40 AM – 02:35 PM, Thursday: 11:40 AM – 02:35 PM
Consultation Hours:	Tuesday: 01:35 PM – 2.55 PM
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Rationale:	Being one of the fundamental requirements for electrical engineering students, the course emphasizes a good understanding of basic concepts about digital logic circuits. It exploits areas like Boolean algebra, combinational circuits, sequential circuits and memory elements. The students are first taught about the number system and logic gates before introduction to digital IC technology. It will be followed by implementation of Verilog code in FPGA board. The aim of the course is to familiarize students with modern design methodology to illustrate how digital design is carried out in practice today.
Pre-requisite (if any):	N/A
Objectives:	
1. Basic knowledge on logic gate implementation and get familiar with IC.	
2. Understanding different combinational circuits and design circuits.	
3. Able to solve problems using sequential circuit and logic design.	

Textbook

1. Digital Systems-Principles and Application- Tocci (10th Edition)
2. Digital Fundamentals- Floyd (8th Edition)

Reference Books

1. Digital Logic Design – Morris Mano (5th Edition)

*** Additional reading material: PowerPoint Slides & PDFs provided by the teachers**

Assessment vs. CO/WA (PO)/Bloom's Taxonomy Level/WP/EA Mapping

I. Course Information

CO	CO Statements	PO	KP	WP	EA	Bloom's Level	Delivery methods and activities	Assessment Tools
CO1	Basic knowledge on logic gate implementation and get familiar with IC.	2	3 5	1 3	1	C1, C2, C3, P3, A2	Lectures, Lab Manual, PPT, PDF White Board Writing	Sessional Class Participation/ Observation Lab Reports, Quiz Test, Lab Project, Lab Final, Viva Voce
CO2	Understanding different combinational circuits and design circuits.	3				C3, C6, ,P3		
CO3	Able to solve problems using sequential circuit and logic design.	5				C3, C5, C6, P3, P4		

Legends: CO-Course Outcomes PO-Program Outcomes, KP-Knowledge Profile
Bloom's Level: Cognitive Domain (C1-C6), Psychomotor Domain (P1-P7), and Affective Domain (A1-A5)

Mapping of Course Outcomes (CO) and Program Outcomes (PO):

COs	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1		H										
CO2			H									
CO3					H							

High (H) = 3, Medium (M) = 2 and Low (L) = 1

II. Assessment vs. CO/PO/BL/WP/EA Mapping

Lab test/Project, Written test/Quiz, Viva Voce, Attendance, Data Sheet/Home Assignments/Report, Class Performance /Continuous Evaluation

Assessment Methods	Description	Grading	CO	Blooms Taxonomy
Final Evaluation	Lab test	30%	CO3	C3, C5, C6, P4
Mid Term Evaluation	Lab Test	20%	CO1, CO2	C1, C2, A2
Viva-Voce/Presentation Skills	Viva Voce	10%	CO1	C1, C2
Report Writing	Data Sheet/Home Assignment/ Report	15%	CO2, CO3	C6, P4
Conduct of Lab test/Lab Performance	Lab Performance	25%	CO1	C1, C2, C3, P3, A2
Total Marks		100%		

Sessional Lecture Plan

Sl. No.	Content of Course	Lecture Hours	COs	Teaching Aids	Teaching Assessment Tools
01	Introduction to different equipment, primary and universal Logic Gates.	03 Hrs.	CO1	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
02	Design and Implementation of Logic Gates using integrated circuits (ICs).	03 Hrs.	CO1	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
03	Design and Implementation of Logic Gates using diode and transistor.	03 Hrs.	CO1	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
04	Design and Implementation of Boolean function using diode and transistor.	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
05	Design and Implementation of Boolean Function using Boolean algebra and K-map.	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
06	Design and Implementation of 2 bit and 3-bit Adder and Subtractor.	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
07	Midterm Examination	03 Hrs.	CO1 CO2		

08	Design and implementation of different Latch and SR Flip Flop.	03 Hrs.	CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
09	Design and implementation of D, T and JK Flip Flop using transistor kit.	03 Hrs.	CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
10	Design and implementation of different shift registers.	03 Hrs.	CO2 CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
11	Design and implementation of Encoder and Decoder.	03 Hrs.	CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
12.	Design and implementation of Multiplexer and De Multiplexer.	03 Hrs.	CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
13.	Design and implementation of DAC and ADC.	03 Hrs.	CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion
14.	Lab Final Assessment	03 Hrs.	CO1		

Teaching and Learning Strategies (TLS)	
TLS1	Delivery lectures using multimedia of different topics
TLS2	Active and fruitful discussion in class regarding efficient solving of the logical, mathematical and complex engineering problems
TLS3	Group discussion and presentation regarding diverse problems and corresponding lectures
TLS4	Evaluation of class performance to reach each student in a class for every topic.

***Additional reading material:** PowerPoint slides & PDF provided by the teacher

Minimum Attendance:

80% class attendance is mandatory for a student in order to appear at the final examination.

tudent's Responsibilities:

1. Students must come to the class prepared for the course material covered in the previous class
2. They must submit their assignments on time
3. No late or partial assignments will be acceptable

Grading Policy:			
Numerical Score	**Letter Grade**		**Grade points**
80% and above	A+	A (Plus)	4.00
75% to > 80%	A	A (Regular)	3.75
70% to > 75%	A-	A (Minus)	3.50
65% to > 70%	B+	B (Plus)	3.25
60% to > 65%	B	B (Regular)	3.00
55% to > 60%	B-	B (Minus)	2.75
50% to > 55%	C+	C (Plus)	2.50
45% to > 50%	C	C (Regular)	2.25
40% to > 45%	D	D (Pass)	2.00
below 40%	F	Fail	0.00
Incomplete	I	-	-
Withdrawal	W	-	-
Project / Thesis Continuation	X	-	-

Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) represent the knowledge, skills and attitudes the students should have at the end of a four years engineering program. CSE program of BAUET has 12 Program Outcomes. They are briefly described in the following table.

PO	Category	Description
PO1	Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems
PO2	Problem Analysis	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
PO3	Design creative solutions	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as

		well as resource, cultural, societal, and environmental considerations as required (WK5).
PO4	Investigation	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5	Modern Tool Usage	Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
PO6	The Engineer: Society, Economy, Environment and Sustainability	When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).
PO7	Ethics	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).
PO8	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9).
PO9	Communication	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences
PO10	Project Management	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team and to manage projects and in multidisciplinary environments
PO11	Life Long Learning	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8).
PO12	Entrepreneurship	Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.

Knowledge and Attribute Profile (KPs)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Range of Complex Engineering Problem Solving (Need High Taxonomy Level)

Attribute: Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7

WP1	Depth of Knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
WP2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
WP3	Depth of analysis required	Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
WP4	Familiarity of issues	Involve infrequently encountered issues or novel problems
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice for professional engineering

WP6	Extent of stakeholder involvement and conflicting requirements	Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
WP7	Interdependence	Address high level problems including many components or sub-problems that may require a systems approach

Range of Complex Engineering Activities (Project based)

Attributes: Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:

EA1	Range of resources	Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
EA2	Level of interaction	Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
EA3	Innovation	Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
EA4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
EA5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches

Bloom's Taxonomy Domain:

Cognitive Domain		**Psychomotor Domain**		**Affective Domain**	
C1	Remembering	**P1**	Perception	**A1**	Receive
C2	Understanding	**P2**	Set	**A2**	Respond
C3	Applying	**P3**	Guided Response	**A3**	Value
C4	Analyzing	**P4**	Mechanism	**A4**	Organize
C5	Evaluating	**P5**	Complex Overt Response	**A5**	Internalize
C6	Creating/Designing	**P6**	Adaption		

Instructions for the students to conduct Sessional Class

SL	**Instructions**
1.	Experiment list for the sessional has been displayed. Each experiment is given a number.

2.	For each day, every student must have to implement the given experimental task according to experiment number. Marks will be given by the faculty for the implemented experiment.
3.	Students may be permitted to carry out missed experiments in their own time with the approval of faculty only.
4.	Each sessional report must be submitted within 15 minutes after starting the class time by wrapping a cover file. Report must be hand written format. Each report will contain the following parts- a. Experiment No. b. Experiment Name c. Theory d. Equipment/ Apparatus e. Circuit Diagram f. Observation Table/ Truth Table g. Discussion
5.	Sessional marks awarded at the end of the semester will be strictly followed according to the sessional assessment method.

Details Sessional Plan:

Experiment No: 01

Experiment Name: Introduction to different equipment, Primary and Universal Logic Gates.

SL	Assigned Tasks	CO
1.	**Objectives of the Lab:** 1. To become familiar with basic digital electronics lab equipment such as breadboards, logic probes, IC testers, and power supplies. 2. To identify and understand the functionality of **primary logic gates** (AND, OR, NOT). 3. To explore the working of **universal logic gates** (NAND, NOR) and understand how they can be used to construct any other logic gate. 4. To practice circuit implementation using logic gate ICs on a breadboard. 5. To verify the truth tables of primary and universal gates through hands-on experimentation. **Learning Outcomes:** After completing this lab, students will be able to: 1. Identify and operate standard digital lab equipment confidently. 2. Describe the operation and truth tables of basic logic gates (AND, OR, NOT) and universal gates (NAND, NOR). 3. Implement simple logic circuits using primary and universal gates. 4. Demonstrate the use of universal gates to replicate other logic gates. 5. Build, test, and troubleshoot basic logic circuits on a breadboard.	C01

Experiment No: 02

Experiment Name: Design and Implementation of Logic Gates using integrated circuits (Ics).

SL	Assigned Tasks	CO
1.	**Objectives:** • To familiarize students with basic digital lab equipment and integrated circuits. • To understand the working principles and truth tables of primary (AND, OR, NOT) and universal (NAND, NOR) logic gates. • To design and implement logic gate circuits using ICs and verify their outputs practically. **Learning Outcomes:** By the end of this lab, students will be able to: • Identify and use essential lab tools and digital ICs. • Implement and test primary and universal logic gates on a breadboard. • Verify truth tables and construct basic gates using universal gates.	C01

Experiment No: 03

Experiment Name: Design and Implementation of Logic Gates using diode and transistor.

SL	Assigned Tasks	CO
1.	Objectives: • To understand the fundamental working principles of logic gates using discrete components. • To design and implement basic logic gates (AND, OR, NOT) using diodes and transistors. • To analyze the circuit behavior and verify gate functionality through practical observation. Learning Outcomes: By the end of this lab, students will be able to: • Explain how logic gates can be built using diodes and transistors. • Construct and test basic logic gate circuits without using ICs. • Verify the logical behavior of each gate through truth table comparison.	C02

Experiment No: 04

Experiment Name: Design and Implementation of Boolean Function using diode and transistor.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the process of implementing Boolean functions without using simplification techniques. • To design and implement Boolean functions using basic logic gates (AND, OR, NOT). • To analyze and verify the circuit behavior and functionality of the Boolean functions through practical observation. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain how to implement Boolean functions using basic logic gates without simplification. • Construct and test Boolean function circuits without using simplification methods. • Verify the functionality of the Boolean function by comparing the circuit behavior with the truth table.	C02

Experiment No: 05

Experiment Name: Design and Implementation of Boolean Function using Boolean algebra and K-map.

SL	Assigned Tasks	CO
1.	Objectives: • To understand the process of simplifying Boolean functions using Boolean algebra and Karnaugh maps (K-map). • To design and implement simplified Boolean functions based on simplified expressions obtained from Boolean algebra and K-map techniques. • To analyze and verify the circuit behavior and functionality of the simplified Boolean functions through practical observation. Learning Outcomes: By the end of this lab, students will be able to: • Explain the process of simplifying Boolean functions using Boolean algebra and Karnaugh maps (K-map). • Construct and test circuits based on simplified Boolean functions using basic logic gates. • Verify the functionality and correctness of the simplified Boolean functions by comparing the circuit behavior with the truth table.	C02

Experiment No: 06

Experiment Name: Design and Implementation of 2 bit, and 3-bit Adder and Subtractor.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and implementation of binary adders and subtractors. • To design and implement 2-bit, 3-bit, and 4-bit binary adder and subtractor circuits using full adders and full subtractors. • To analyze and verify the behavior of the adder and subtractor circuits through practical observation and truth table comparisons. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the operation of binary adders and subtractors (half adder, full adder, half subtractor, and full subtractor). • Construct and test 2-bit, 3-bit, and 4-bit binary adder and subtractor circuits using basic logic gates. • Verify the functionality of the adder and subtractor circuits by comparing the circuit behavior with the expected truth tables.	C02

Experiment No: 07

Experiment Name: Mid Assessment

SL	Assigned Tasks	CO
1.	There will be a 30-minute practical examination in the 7th week, consisting of a circuit you are expected to build, test, and demonstrate that it works. Every student will work alone on the midterm exam.	C01 CO2

Experiment No: 08

Experiment Name: Design and implementation of different Latch and SR Flip Flop.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and functioning of SR (Set-Reset) Latch and SR Flip Flop. • To design and implement SR Latch and SR Flip Flop circuits using logic gates and flip-flops.	C03

	• To analyze and verify the behavior of the SR Latch and SR Flip Flop circuits through practical observation and truth table comparisons. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the working principles of SR Latch and SR Flip Flop. • Construct and test SR Latch and SR Flip Flop circuits using logic gates and flip-flops. • Verify the functionality of the SR Latch and SR Flip Flop circuits by comparing the circuit behavior with the expected output based on their truth tables.	

Experiment No: 09

Experiment Name: Design and implementation of D, T and JK Flip Flop using trainer kit.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and functioning of D (Data), T (Toggle), and JK Flip Flops. • To design and implement D, T, and JK Flip Flops using logic gates and flip-flops. • To analyze and verify the behavior of these flip-flop circuits through practical observation and truth table comparisons. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the working principles of D, T, and JK Flip Flops. • Construct and test D, T, and JK Flip Flop circuits using logic gates and flip-flops. • Verify the functionality of the D, T, and JK Flip Flop circuits by comparing the circuit behavior with the expected output based on their truth tables.	CO3

Experiment No: 10

Experiment Name: Design and implementation of different shift registers.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and functioning of shift registers, specifically SIPO (Serial In Parallel Out) and PISO (Parallel In Serial Out). • To design and implement SIPO and PISO shift registers using flip-flops and other basic digital components. • To analyze and verify the behavior of the shift register circuits through practical observation and truth table comparisons. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the working principles of shift registers, specifically SIPO and PISO.	CO3

	• Construct and test SIPO and PISO shift register circuits using flip-flops and basic logic gates. • Verify the functionality of the shift register circuits by comparing the circuit behavior with the expected output based on the shift operation.	

Experiment No: 11

Experiment Name: Design and implementation of Encoder and Decoder.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and functionality of encoders and decoders. • To design and implement encoder and decoder circuits using logic gates. • To analyze and verify the behavior of the encoder and decoder circuits through practical observation and truth table comparisons. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the working principles of encoders and decoders. • Construct and test encoder and decoder circuits using basic logic gates. • Verify the functionality of the encoder and decoder circuits by comparing the circuit behavior with the expected output based on their truth tables.	CO3

Experiment No: 12

Experiment Name: Design and implementation of Multiplexer and Demultiplexer.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and functionality of multiplexers (MUX) and demultiplexers (DEMUX). • To design and implement multiplexer and demultiplexer circuits using logic gates. • To analyze and verify the behavior of the multiplexer and demultiplexer circuits through practical observation and truth table comparisons. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the working principles of multiplexers and demultiplexers. • Construct and test multiplexer and demultiplexer circuits using basic logic gates. • Verify the functionality of the multiplexer and demultiplexer circuits by comparing the circuit behavior with the expected output based on their truth tables.	CO3

Experiment No: 13

Experiment Name: Design and implementation of ADC and DAC.

SL	Assigned Tasks	CO
1.	**Objectives:** • To understand the operation and functionality of Analog-to-Digital Converters (ADC) and Digital-to-Analog Converters (DAC). • To design and implement ADC and DAC circuits using basic digital components. • To analyze and verify the behavior of the ADC and DAC circuits through practical observation and comparison with theoretical values. **Learning Outcomes:** By the end of this lab, students will be able to: • Explain the working principles of ADC and DAC and their role in digital systems. • Construct and test ADC and DAC circuits using basic digital components (such as counters, comparators, etc.). • Verify the functionality of the ADC and DAC circuits by comparing the circuit behavior with the expected output and converting values between analog and digital signals.	CO3

Experiment No: 14

Experiment Name: Lab Final Assessment and Quiz.

SL	Assigned Tasks	CO
1.	**There will be a one-hour final assessment.**	C01 CO2 CO3

Approved by

Head of the Department